Aryl hydrocarbon receptor literature

Bibliometric Analysis Summary Report

Example configuration for a conservative OpenAlex bibliometric pipeline focused on the aryl hydrocarbon receptor literature.

Query definition

OpenAlex exact-match title/abstract retrieval for "aryl hydrocarbon receptor", "ah receptor", and "dioxin receptor", plus title-level "AHR" hits; local regex validation retained records with explicit AhR naming in the title or abstract-phrase hits supported by biologically relevant title language.

Key corpus statistics

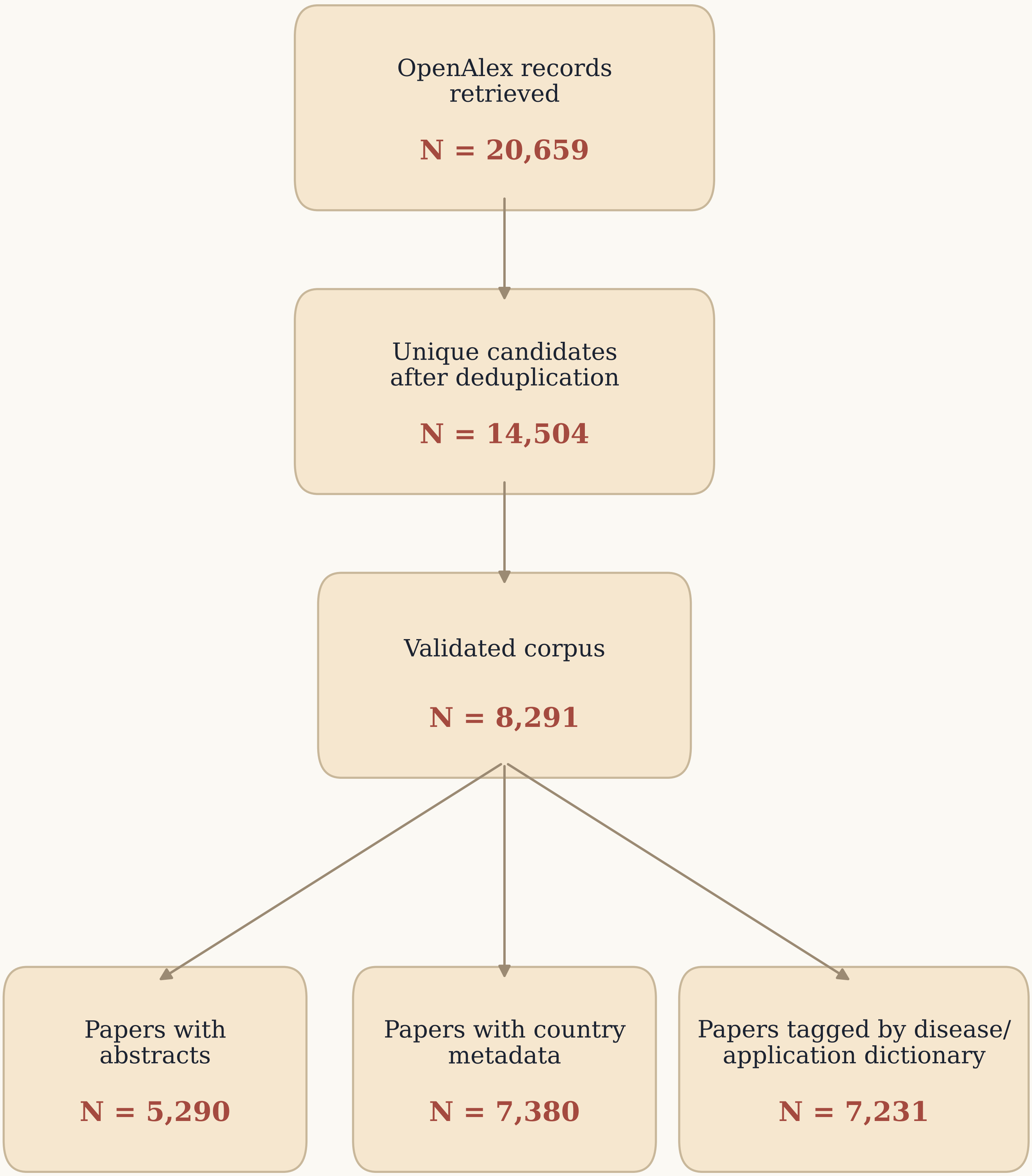
Validated papers	8,291
Time window	1980 to 2026
Abstract coverage	5,290 (63.8%)
Country metadata	7,380 (89.0%)
Disease/application tagged	7,231 (87.2%)
Focus-tagged papers	6,766 (81.6%)
Journals represented	1,609
Countries represented	107

Report scope

The following pages reproduce the generated thesis figures in order. This report is assembled from the current figure files without altering their styling or content.

Corpus Filtering and Retention Flow

Technical overview of retrieval, deduplication, validation, and metadata availability.



Editable Mermaid source is written alongside this PDF for reuse in thesis documentation.

Figure 00. Corpus filtering and metadata retention flow

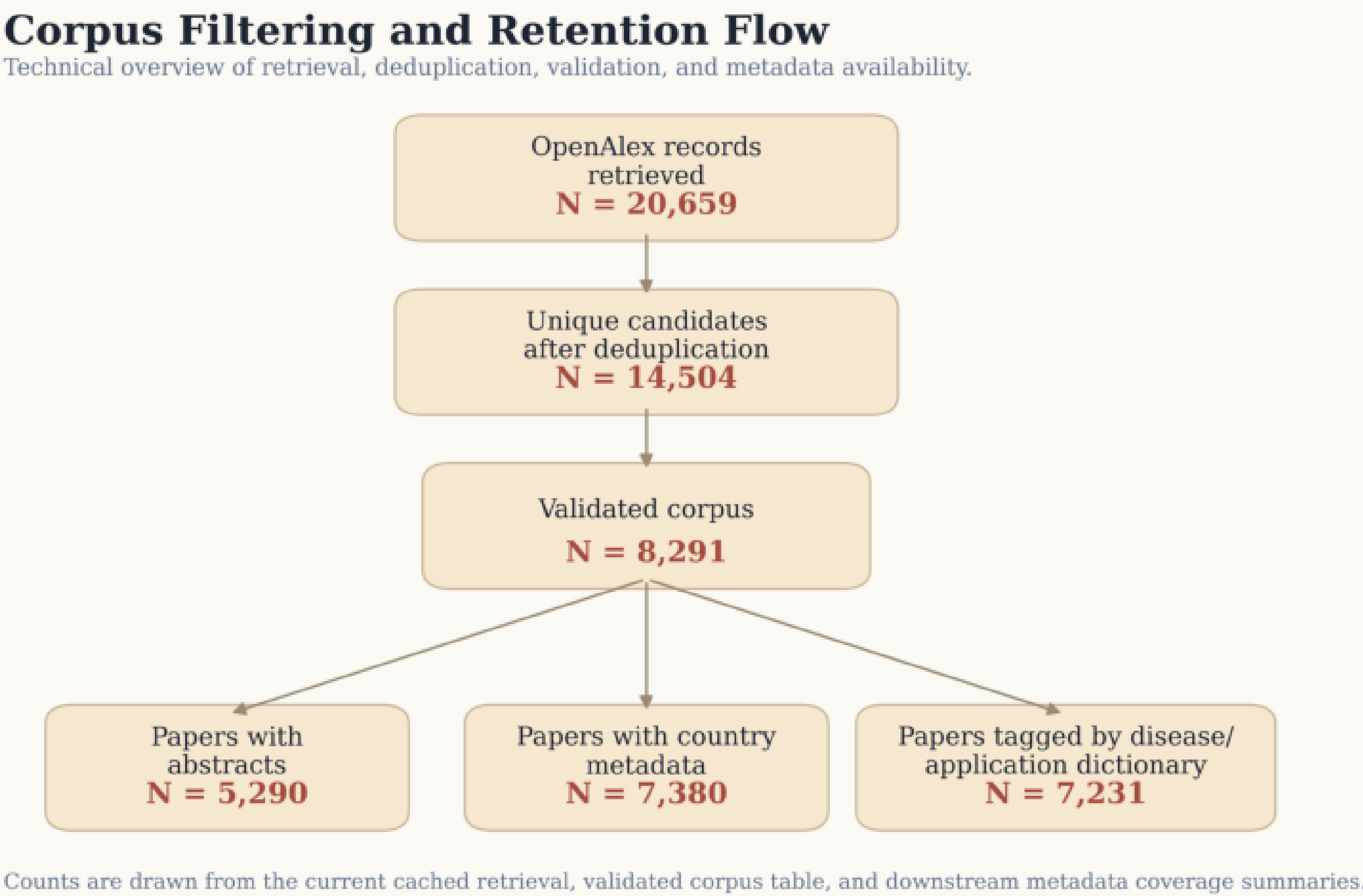


Figure 01. AhR literature growth over time

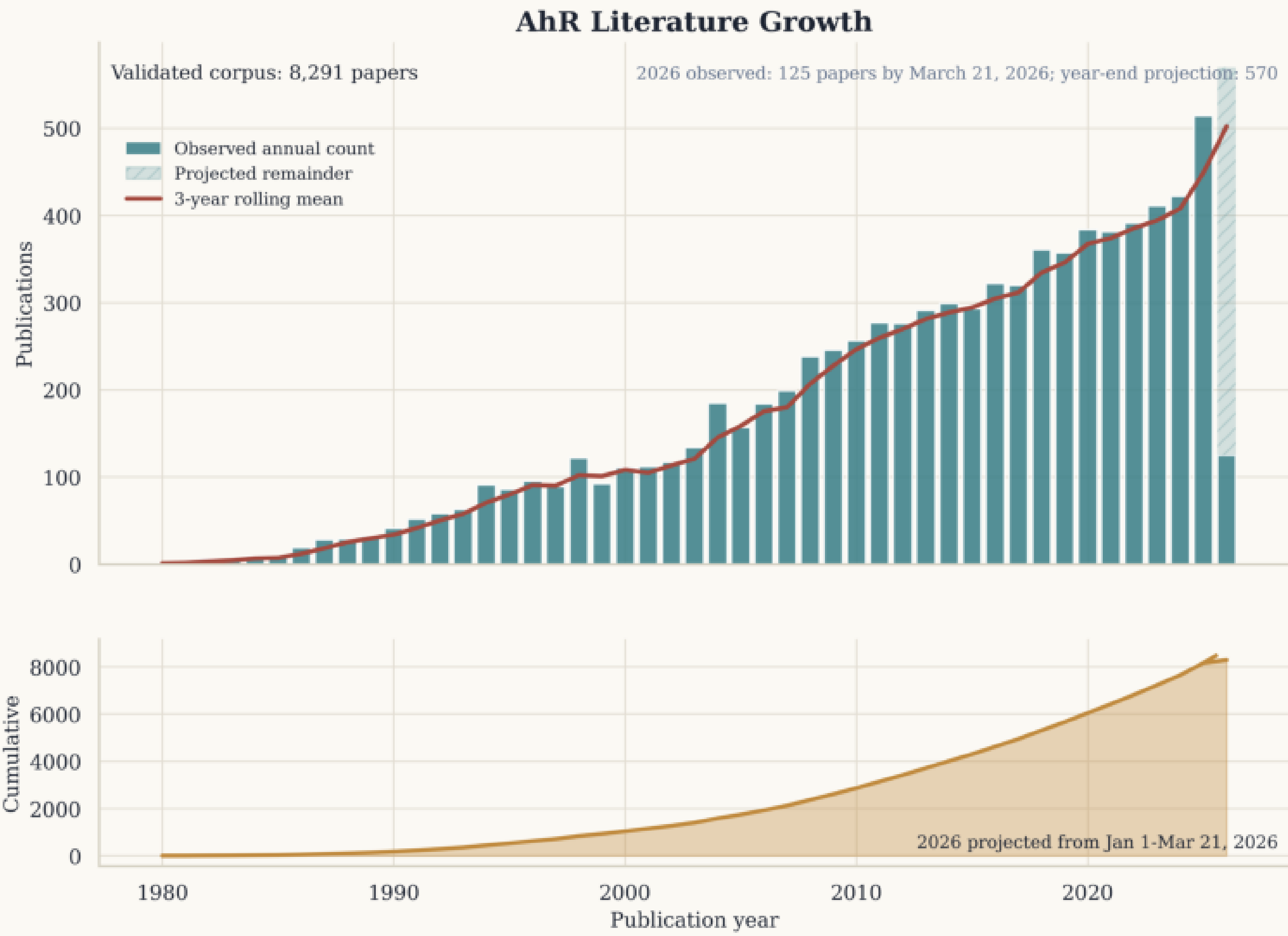


Figure 02. Disease and application distribution across the AhR corpus

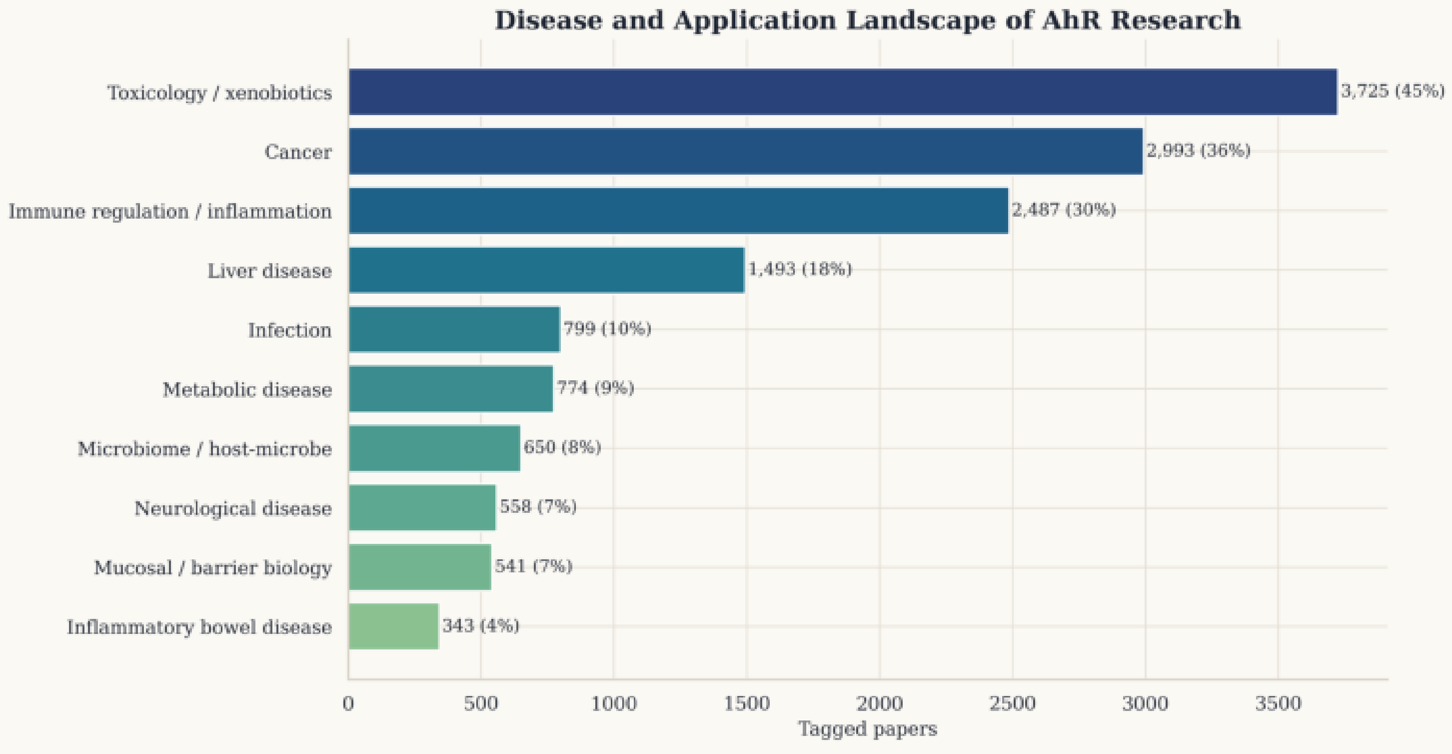
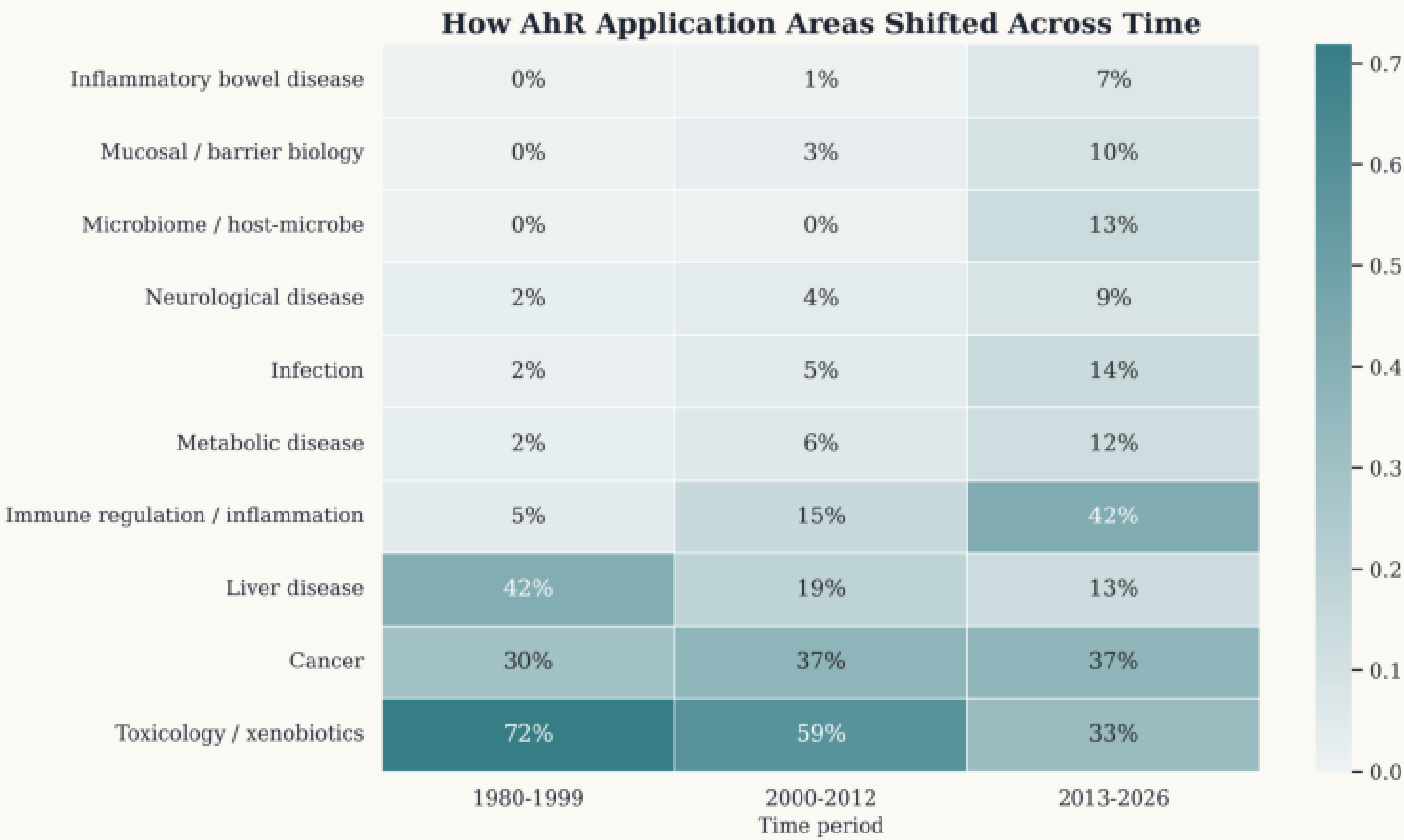
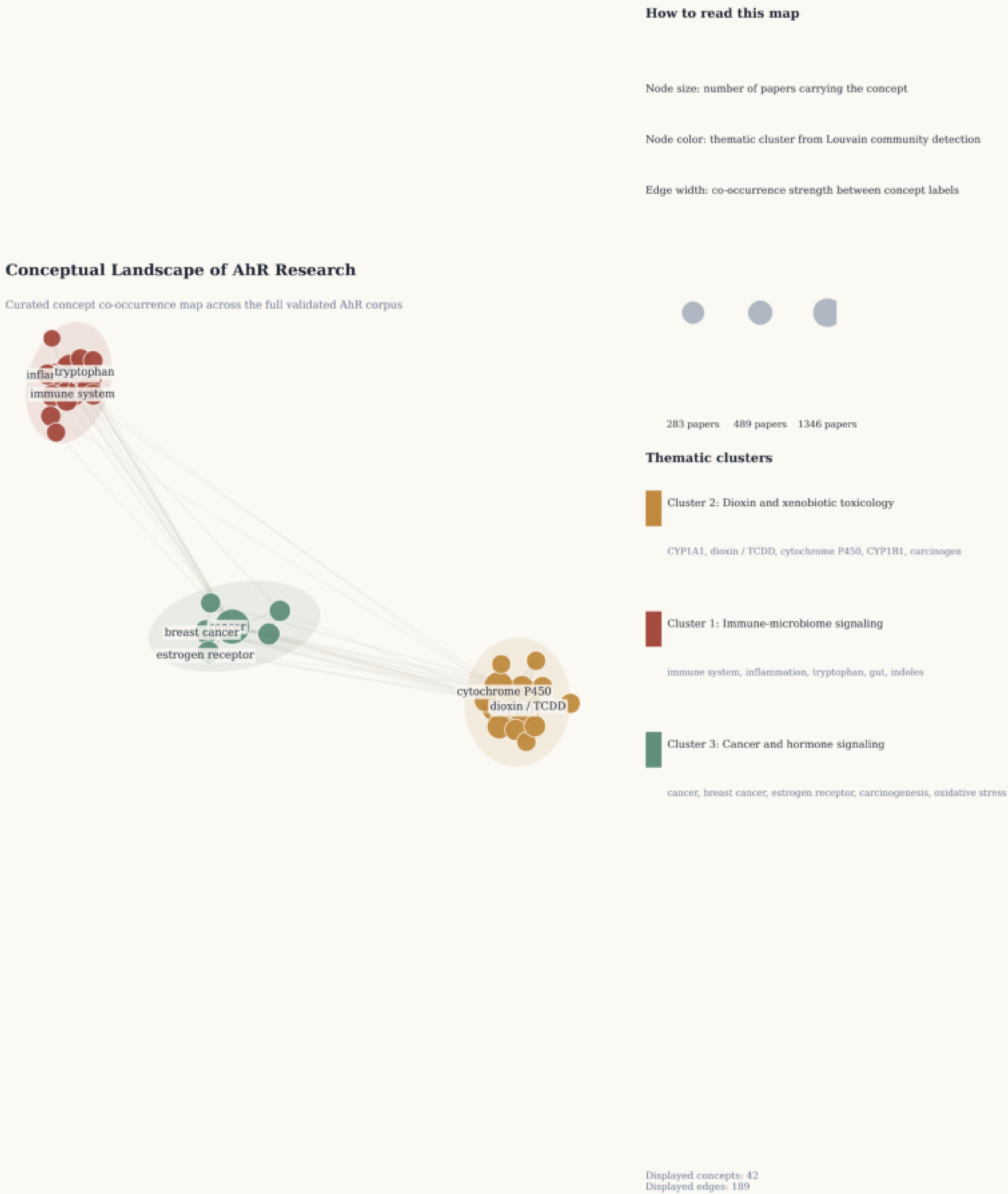


Figure 03. Disease and application trends across AhR field eras



Conceptual Landscape of AhR Research



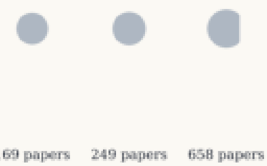
Immune-Microbiome-Barrier AhR Sublandscape

How to read this map

Node size: number of papers carrying the concept

Node color: thematic cluster from Louvain community detection

Edge width: co-occurrence strength between concept labels



Thematic clusters

Cluster 1: T cell and cytokine regulation

inflammation, immune system, Treg cells, TNF-alpha, Th17 cells

Cluster 2: CYP1 toxicology and carcinogenesis

cancer, CYP1A1, dioxin / TCDD, cytochrome P450, CYP1B1

Cluster 4: Microbiome and gut barrier biology

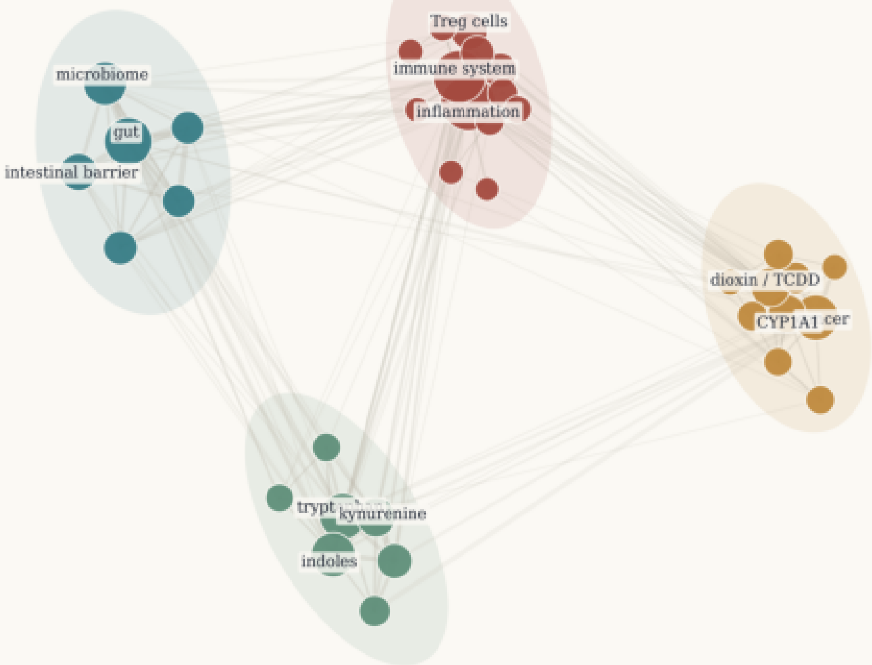
gut, microbiome, intestinal barrier, IL-22, inflammatory bowel disease

Cluster 3: Tryptophan-kynurenine immunometabolism

tryptophan, indoles, kynurenine, IDO1, tumor immunity

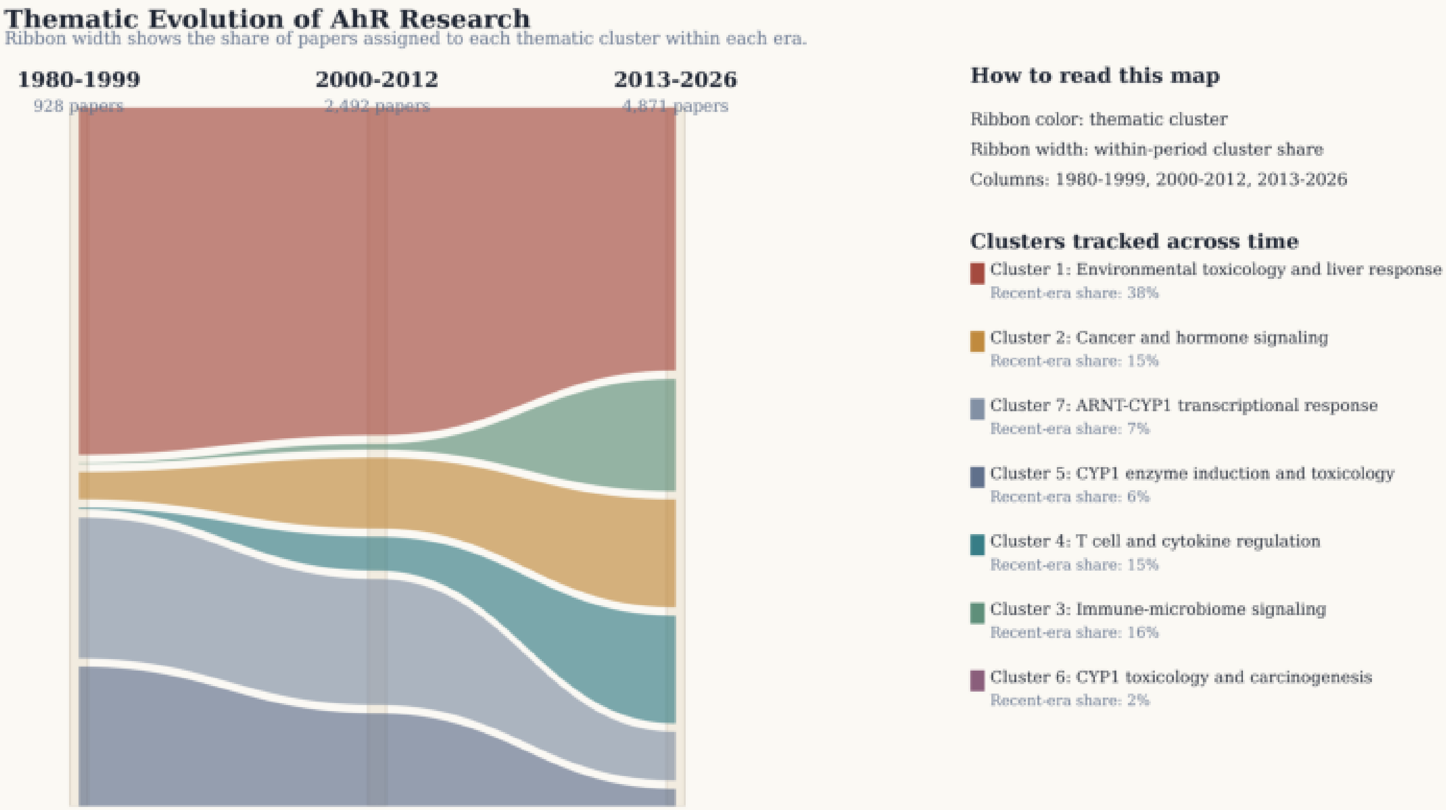
Immune-Microbiome-Barrier AhR Sublandscape

Focused concept map of the microbiome, gut, mucosal, inflammatory, and immunoregulatory AhR literature



Displayed concepts: 36
Displayed edges: 203

Thematic Evolution of AhR Research



Document Landscape of AhR Research Themes

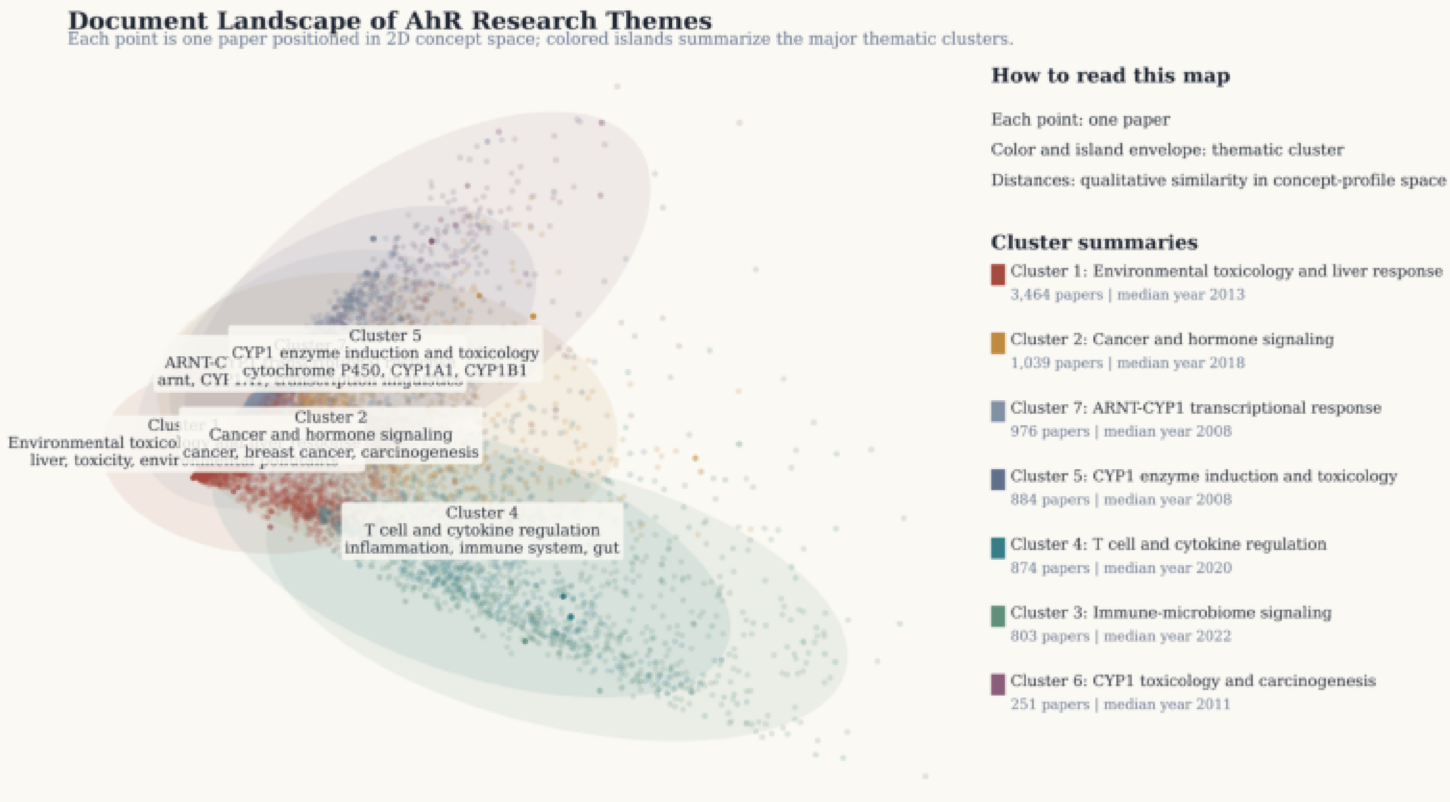
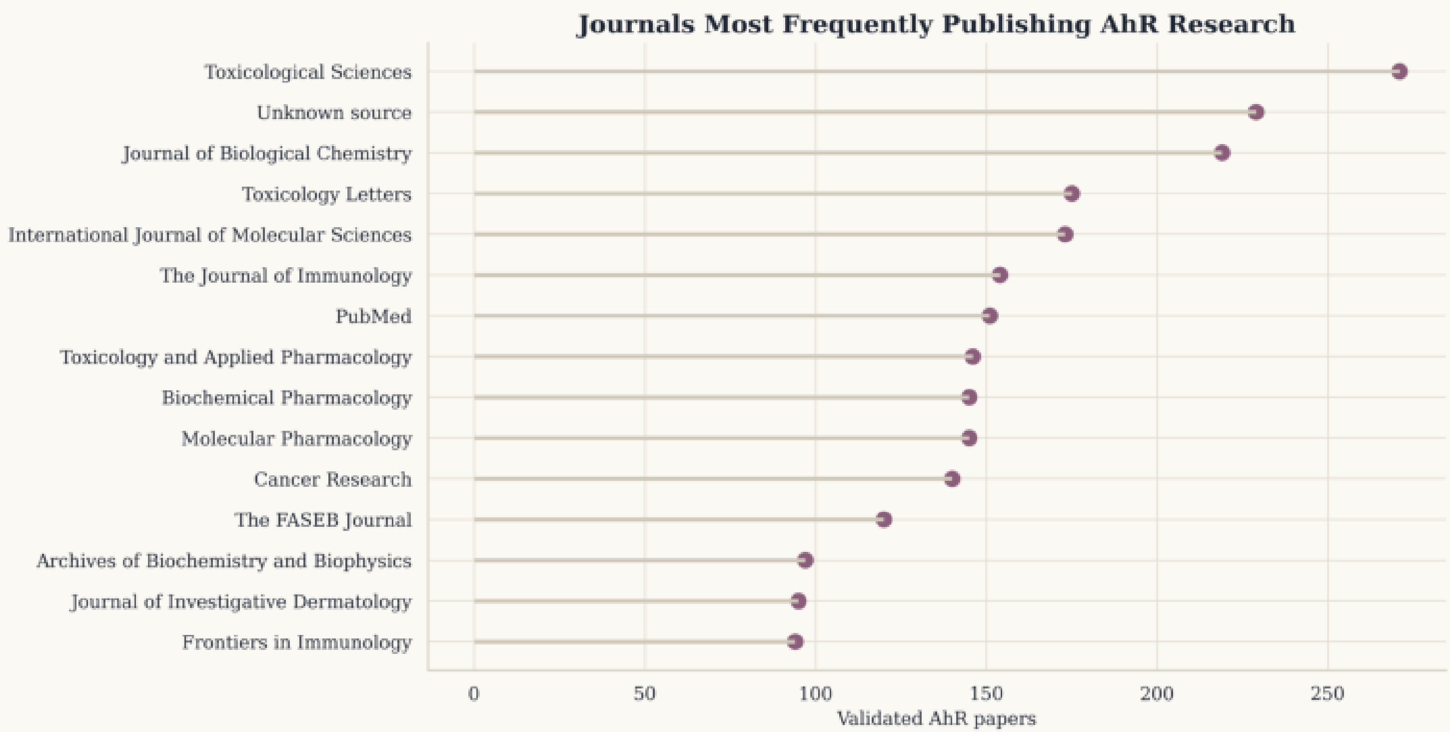
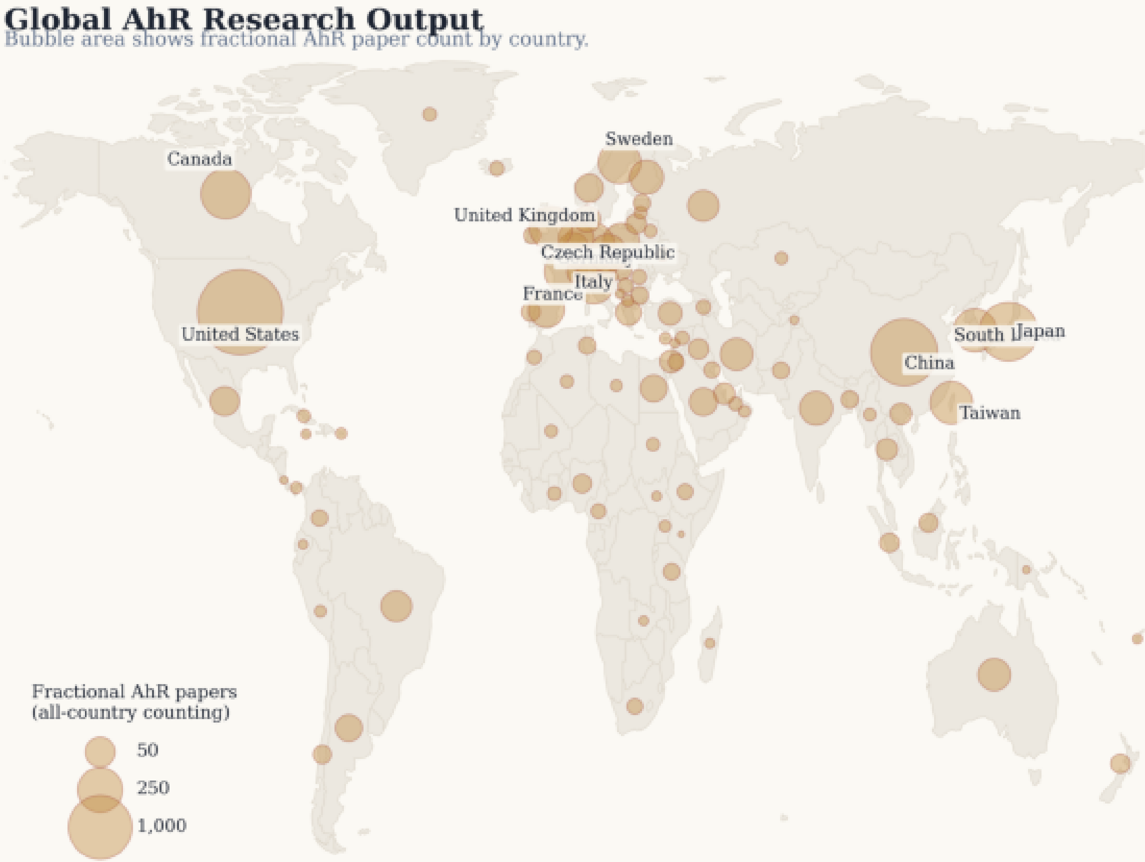


Figure 08. Journals most frequently publishing AhR papers

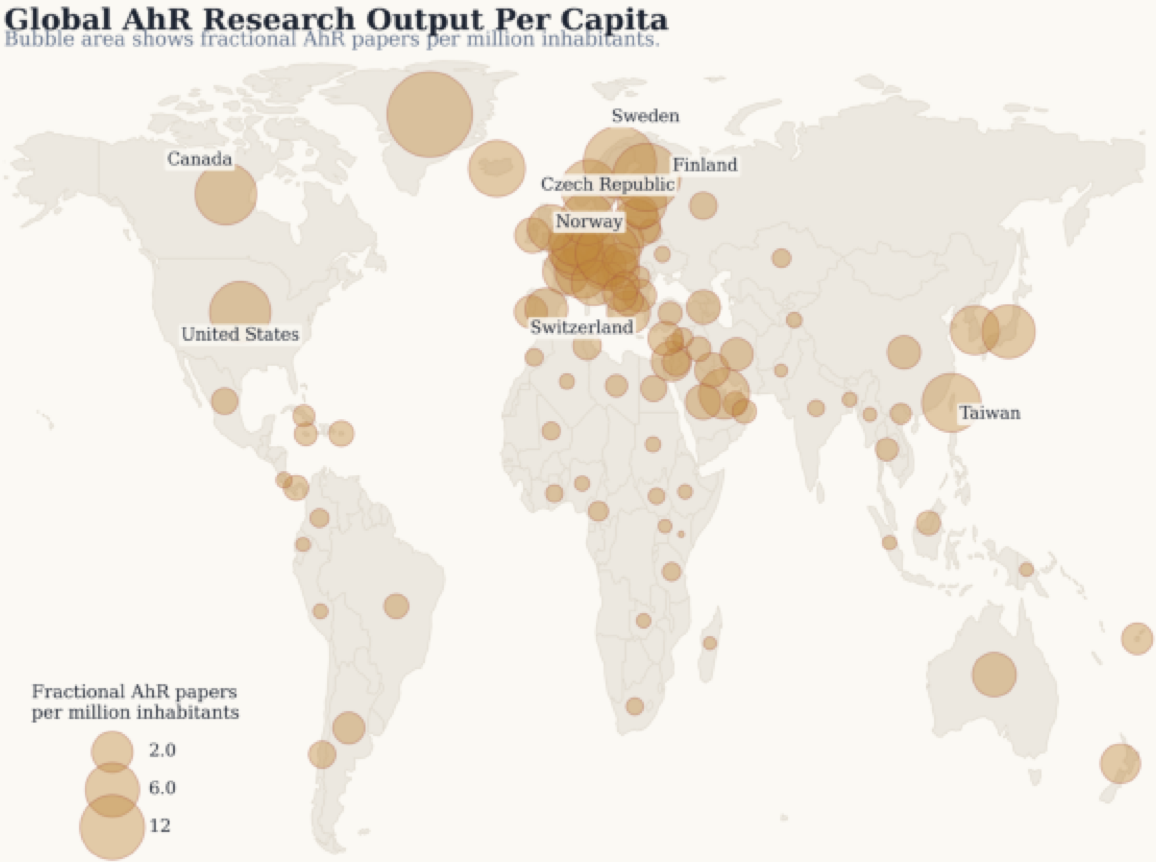


Global AhR Research Output



Fractional counting distributes each paper across all represented author-affiliation countries. Country metadata available for 7,380 of 8,291 papers (89%).

Global AhR Research Output Per Capita



Fractional counting distributes each paper across all represented author-affiliation countries. Country metadata available for 7,380 of 8,291 papers (89%).